Introduction to Error Analysis

1 Introduction

Physics is an observational discipline, where we try to measure quantities of interest to us (dependent variables) while varying parameters (independent or control variables) that we can control. Based on our observations, we attempt to develop models to explain why the measured quantities depend on the control variables in the observed ways. In order for this process to work, we need to identify and quantify the uncertainty in our measured values. For example, imagine that we have a ruler that has markings spaced by 1 cm. With such a ruler, we clearly cannot measure a length with a precision of 1 mm. In general, finite instrument resolution, inherent noise in our equipment, and other non-ideal aspects of how our experiment is set up, all contribute to an uncertainty in our final measurement. Ultimately, the reliability of the models we build depends on how accurately and precisely we have measured our data. Hence as experimenters, we must always report an uncertainty in our measurement that is quantitatively determined taking into account the various limitations of our experiment. This process of quantitatively determining the uncertainty in our measured values is called error analysis. In this document, we will learn how to identify some common types of errors in our measurements and quantify them, and how to propagate errors when we have to combine multiple measured values to determine our quantity of interest.

2 A model setting

As a concrete example, we will consider measuring the acceleration due to gravity g by measuring the time period of a simple pendulum. We know that the time period of a simple pendulum of mass m is given by $T = 2\pi\sqrt{L/g}$ where L is the length of pendulum. Hence, we can determine the value of g as

$$g = 4\pi^2 \frac{L}{T^2}. \tag{1}$$

if we measure the values of L and T. However, we can only measure these quantities to a finite uncertainty and we have to account for this uncertainty in our knowledge of these quantities when quoting the final value of g. Let us now discuss the uncertainties in L and T, before moving on to discuss how we can combine these uncertainties to obtain the error in our measurement of g.

3 Least count error

Let us suppose that we are measuring the length of the pendulum using a metre scale with divisions of 1 mm. In other words, the least count of the scale is 1 mm. We measure it and find that, keeping the point of suspension at the 0 mark, the centre of the bob falls between 60.1 cm and 60.2 cm. Because of a lack of further resolution, all we can say is that the length of the pendulum is somewhere in the interval $[60.1\text{cm}, 60.2\text{cm}]$. It is reasonable to assume that, given

this observation, any value in this interval has an equal probability of being the true value. For a variable X that is uniformly distributed in an intervale $[a, b]$, the variance of X, denoted as $(\Delta X)^2$ is given by

$$(\Delta X)^2 = \frac{(b-a)^2}{12}. \tag{2}$$

Taking the interval $[60.1\text{cm}, 60.2\text{cm}]$, the standard deviation of a variable that is uniformly distributed in this interval is $\sqrt{(0.1)^2/12} \approx 0.03$ cm. Hence, the measured result can be quoted as

$$L = \overline{L} \pm \sigma_L = 60.15 \pm 0.03 \text{ cm}, \tag{3}$$

where we take the mid-point of the interval as our best estimate and the standard deviation of the uniform distribution in the interval as the uncertainty in our result.

The above is an example of how least count of the instrument leads to an uncertainty in our measured result. In short, if the least count of the instrument is c, then $c/\sqrt{12}$ corresponds to the uncertainty in our measured reading.

Note that, while using vernier calipers, we do not take the mid-point of two adjacent divisions as our reading, but instead we take the reading as

$$l_{\text{read}} = (\text{main scale reading}) + (\text{vernier coincidence}) \times (\text{least count}). \tag{4}$$

In this case, the uncertainty is still $c/\sqrt{12}$ where c is the least count of the instrument. In the case of the metre scale, there is no such coincidence procedure offered by the instrument and hence our best estimate is just the mid-point of the two divisions between which our observed length falls. Hence, care must be taken to quote the central value of our measurement, which depends on how the instrument is supposed to be used for taking readings.

4 Statistical error

Let us suppose that we measure the time period of the pendulum by allowing it to oscillate, say, 10 times, and measuring the total elapsed duration using a digital timer. Dividing this duration by 10 gives us one estimate t_1 of the time period T. Firstly, note that it is advisable to measure a time period using multiple complete oscillations instead of just one as it will average out random errors we might make in our measurements over just one period, such as in our visual estimate of the start and end times of an oscillation. On top of this, it is advisable to repeat our measurement multiple times, i.e., we can let the pendulum oscillate again for 10 more cycles, and get a second estimate t_2 of the time period T. In general, we can make N such measurements of the time period. We can then arrive at an estimate for the time period as the average of these N measurements:

$$\overline{T} = \frac{t_1 + t_2 + \ldots + t_N}{N}. \tag{5}$$

The quantity we have computed is called a sample mean. Assuming we did each repetition in as identical a manner as possible, it is reasonable to assume that the varied outcomes $t_1, t_2, \ldots, t_N$ are all independently drawn from the same underlying probability distribution. The outcomes are probabilistic because of imperfections such as human error in estimating start and end times of oscillations, timing jitters in our timer, etc. Under this assumption that all the t_i's are independent and identically distributed (i.i.d.), the error in the mean value, T, is given by

$$(\Delta T)^2 = \frac{\sigma^2}{N}, \tag{6}$$

where σ is the standard deviation of the probability distribution from which each t_i is drawn. This result follows simply from the well-known transformations of the variance when adding two or more independent variables and when scaling a variable by a constant ($1/N$ in this case). As we can see, the statistical error/uncertainty goes down with increasing number of repetitions. Hence, it is good to repeat noisy measurements as many times as possible to reduce the statistical error.

However, a practical problem is that we often do not know what distribution is followed by our measurement readings, since we do not have a good model of the various processes contributing to the statistical fluctuations. In this case, we can use our repeated measurements to construct a quantity called the sample variance, which serves as an estimate of the true variance of the underlying probability distribution. The sample variance, denoted s^2 here, is given by

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(t_i - \overline{T})^2. \tag{7}$$

Note that the factor in the denominator is $N-1$ instead of N. We will take this result for granted here, although it can be proved rigorously why this should be the case. The intuition is that although we made N observations, one 'degree of freedom', namely, the average of the N measurements, was already used to estimate the mean, so we only have $N-1$ independent effective data points to estimate the variance. Alternatively, it is reasonable that with just $N = 1$ measurement, we cannot estimate both the mean and the fluctuations about the mean, so the sample variance must be undefined for $N = 1$, which is the case in the above formula.

Finally, we can quote our measured value for the time period as

$$T = \overline{T} \pm \sigma_T, \tag{8}$$

where

$$\sigma_T = \frac{s}{\sqrt{N}}. \tag{9}$$

5 Error propagation

We now have a measurement of the length L and the time period T along with their uncertainties. Let us discuss how we can determine the uncertainty in the quantity g, which is what we wish to measure. Suppose a quantity u is a function of multiple variables $x, y, \ldots$, i.e., $u = f(x, y, \ldots)$. Let us suppose we have measured each of these quantities independently, for example, x could be the length and y could be the time period. So, our estimate of x is $\overline{x} \pm \sigma_x$, where σ_x is the uncertainty in x and so on for the other quantities. Let us consider a small fluctuation δx, δy, ... in the variables about their average values $\overline{x}, \overline{y}, \ldots$. Then, Taylor expanding both sides up to first order, we have

$$\overline{u} + \delta u = f(\overline{x} + \delta x, \overline{y} + \delta y, \ldots) = f(\overline{x}, \overline{y}, \ldots) + \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y + \ldots, \tag{10}$$

where the partial derivatives are evaluated at $\overline{x}, \overline{y}, \ldots$. Then, we see that

$$\overline{u} = f(\overline{x}, \overline{y}, \ldots). \tag{11}$$

To find the uncertainty in u, we have to estimate the size of the fluctuation δu. Its mean value is zero because the mean value of the fluctuation $\delta x, \delta y$ are zero. However, this does not mean there is no fluctuation, instead we have to find the root-mean-square or RMS size of the

fluctuation. This is because the quantities σ_x^2, σ_y^2 are the mean values of the squared deviations $(\delta x)^2, (\delta y)^2$ and so on. Hence, squaring the fluctuations on both sides, we have

$$(\delta u)^2 = \left(\frac{\partial f}{\partial x}\right)^2 (\delta x)^2 + \left(\frac{\partial f}{\partial y}\right)^2 (\delta y)^2 + 2\left(\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}\right)(\delta x \delta y) + \ldots. \tag{12}$$

We assume that the fluctuations in $x, y, \ldots$ are all uncorrelated. In the example of the simple pendulum experiment, this is reasonable because the uncertainty in the length measurement has a different origin than the uncertainties in the time measurement. Hence, taking an average on both sides and setting the average values of the cross-terms to 0, we arrive at

$$\sigma_u^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + \ldots. \tag{13}$$

This result is the quantitative statement of the fact that independent errors 'add in quadrature' (like the sides of a right triangle in the Pythagoras theorem).

Returning to the example of the simple pendulum, our estimate of g is just given by

$$\overline{g} = 4\pi^2 \frac{\overline{L}}{\overline{T}^2}. \tag{14}$$

The uncertainty requires us to evaluate the partial derivatives:

$$\frac{\partial g(L,T)}{\partial L} = 4\pi^2 \frac{1}{\overline{T}^2}, \quad \frac{\partial g(L,T)}{\partial T} = -8\pi^2 \frac{\overline{L}}{\overline{T}^3}. \tag{15}$$

Thus, the uncertainty σ_g is computed as

$$\sigma_g^2 = \left(4\pi^2 \frac{1}{\overline{T}^2}\right)^2 \sigma_L^2 + \left(8\pi^2 \frac{\overline{L}}{\overline{T}^3}\right)^2 \sigma_T^2. \tag{16}$$

The final result is then reported as

$$g = \overline{g} \pm \sigma_g. \tag{17}$$

6 Combining multiple estimates with unequal uncertainties

Let us now suppose that we have carried out the simple pendulum experiment for different suspension lengths $L_1, L_2, \ldots L_K$. We can then go through the above analysis and arrive at M estimates of the value of g: $\overline{g}_1 \pm \sigma_{g,1}, \overline{g}_2 \pm \sigma_{g,2}, \ldots, \overline{g}_K \pm \sigma_{g,M}$. Clearly, we have more information about the value of g from multiple experiments than from a single experiment. How can we combine these different estimates to obtain a more precise estimate?

Naively, we could think of taking an average of the K estimates. However, different estimates have different uncertainties. Intuitively, we would rely more heavily on the estimate with lower uncertainty than one with a larger uncertainty. According to this intuition, then, taking a simple average cannot be the right procedure as we give equal importance to all estimates regardless of the uncertainty. It turns out that the correct way to combine multiple independent estimates is to take a weighted average (see equations below), where the weight assigned to each estimate is the reciprocal of the squared uncertainty. That is, choose weights $w_i = C/\sigma_i^2$, where C is a normalization constant such that $\sum_i w_i = 1$. After solving for C, we find the weights to be

$$w_i = \frac{1/\sigma_i^2}{\sum_{k=1}^{M} \frac{1}{\sigma_k^2}}. \tag{18}$$

The combined optimal estimate can then be found to be

$$\overline{g} \pm \sigma_g, \tag{19}$$

where

$$\overline{g} = \sum_{i=1}^{M} w_i \overline{g}_i, \tag{20}$$

and

$$\sigma_g^2 = \frac{1}{\sum_{k=1}^{M} \frac{1}{\sigma_{g,k}^2}}. \tag{21}$$

If all the K independent estimates have equal uncertainty, then we see that the weights all become equal to $1/M$, and the weighted average is just a simple average. The resulting uncertainty (variance) in the averaged quantity is again reduced by a factor M. In contrast, when the uncertainties are unequal, the more precise estimates contribute with higher weight than the ones with larger uncertainties. In other words, we 'rely' more heavily on the more precise estimates, which is line with our intuitive expectation.

In fact, the above analysis is just an extension of the error analysis for statistical error under repeated measurements, except that each measurement now has a known but different uncertainty. In the case of the time period measurement, we had assumed that each measurement had the same uncertaintly, but we did not know what it was beforehand. Instead, we estimated it using the sample variance.

7 Systematic Error

Finally, we briefly mention the important class of errors called systematic errors. These are errors that arise due to problems in the design of an experiment. As a simple example, if an experimenter always reads a metre scale at a particular oblique angle instead of vertically, then all the readings suffer from a parallax error that shift them all in the same direction. As a second example, if there is a constant off-set in an instrument, such as a voltmeter that has not been accounted for, all voltage readings are shifted by a constant amount. If the source of systematic error can be identified and its effect corrected for, then it must be done before the experiment is performed. Often, the presence of systematic errors can be inferred from the measured data, where for example, a graph that is supposed to be a straight line is obtained to be curved, or a line that is supposed to pass through the origin in theory is experimentally found to have a non-zero intercept. In this lab, you are encouraged to think about the experimental setup and attempt to minimize potential sources of systematic errors if possible. Also, try to diagnose your data at the end of the experiment to see if your readings might have suffered from systematic errors in the experiment. Document your observations regarding potential systematic errors and your attempts at minimizing them in your report.